

A Brief Primer on Theory of Wavelets

Shuvraneel Mitra

May 21, 2026

Abstract

We look at the theory of wavelets, some transforms and then go on to explore multi-resolution analysis.

1 Introduction

Unlike DCTs and DFTs, which use sinusoidal waves as basis functions, this new variety of transformations uses small waves of varying frequency and of limited support, known as wavelets as basis. The wavelets can be scaled and shifted to analyze the spatial frequency contents of an image at different resolutions and positions. A wavelet can therefore perform analysis of an image at multiple resolutions, making it an effective tool in multi-resolution analysis of images. Furthermore, wavelet analysis performs what is known as **space-frequency localization** so that at any specified location in space, one can obtain its details in terms of frequency.

2 Scaling Functions and the functional subspace

Any function can be analyzed in terms of a linear combination of real-valued expansion functions (note that we use the term basis iff the set of functions spans the function space and is orthogonal): $f(x) = \sum_k a_k \varphi_k(x)$ (eg. Fourier series expansion). Let us define a set of transformations, called the *scaling functions*, by scaling and translating a real, square-integrable (but yet unspecified) function $\varphi(x)$:

$$\varphi_{r,s}(x) = 2^{r/2} \varphi(2^r x - s)$$

where $r, s \in \mathbb{Z}$ and $\varphi(x) \in L^2(\mathbb{R})$. s controls the translation in integer steps and r controls the amplitude, as well as the width of the function in the x -direction. By a clever choice of $\varphi(x)$ the set of functions $\{\varphi_{r_0,s}(x)\}$ can be made to cover the entire $L^2(\mathbb{R})$ space.

Now, if we choose any particular scale r_0 , say, the set of functions $\{\varphi_{r_0,s}(x)\}$ obtained through integer translations can only cover a subspace of the entire $L^2(\mathbb{R})$. The subspace so spanned, V_{r_0} , is defined as the functional subspace of $\{\varphi_{r_0,s}(x)\}$ at a given scale r_0 . Since the width of the set of functions $\{\varphi_{r_0+1,s}(x)\}$ is half of that of the set of functions $\{\varphi_{r_0,s}(x)\}$, the latter can be analyzed by the former, but not the other way. Thus, the subspace spanned by the scaling functions at lower scales is contained within the subspace spanned by those at higher scales:

$$V_{-\infty} \subset \cdots \subset V_{-1} \subset V_0 \subset V_1 \subset \cdots \subset V_{\infty}$$

The expansion functions of subspace V_r can be expressed as a weighted summation of the functions of subspace V_{r+1} as follows: $\varphi(x) = \sum_n h_{\varphi}(n) \sqrt{2} \varphi(2x - n)$ where $\varphi(x)$ represents the basic function of V_r and $\varphi(2x - n)$, the basic function of V_{r+1} .

Consider the unit width Haar scaling function

$$\varphi(x) = \begin{cases} 1 & x \in [0, 1) \\ 0 & \text{otherwise} \end{cases}$$

Then, since $\varphi_{0,0}(x) = \varphi(x)$, the subspace V_0 is spanned by the set $\{\varphi_{0,s}(x)\}$.

3 Wavelet Functions

A set of integer translated and binary scaled functions $\{\psi_{r,s}(x)\}$ that **span the difference subspace** between two adjacent scaling-function subspaces is defined as a set of wavelet functions. Consider two consecutive subspaces V_r and V_{r+1} . Then, $\{\psi_{r,s}(x)\}$ will span the subspace $W_r = V_{r+1} \setminus V_r$.

The definitions of the wavelet functions in terms of a base function is again given by

$$\psi_{r,s}(x) = 2^{r/2} \psi(2^r x - s)$$

Then, $V_{r+1} = V_r \oplus W_r$, expanding which we can get

$$L^2(\mathbb{R}) = V_{r_0} \oplus W_{r_0} \oplus W_{r_0+1} \oplus \dots$$

Thus a function belonging to $L^2(\mathbb{R})$ can be defined in terms of a scale- r_0 scaling function and a number of wavelet functions.

Suppose that a function to be analyzed, $f(x)$, belongs to the subspace V_1 but not V_0 . In that case, the scaling functions of V_0 make a crude approximation of $f(x)$ and the wavelet functions of W_0 provide the details. Thus, the scaling functions can be thought of as low-pass filters and the wavelet functions as high-pass filters.

Since wavelet spaces reside within the spaces spanned by the next higher scaling functions, any wavelet function can be expressed as a weighted sum of shifted double-resolution scaling functions as follows: $\psi(x) = \sum_n h_\psi(n) \sqrt{2} \varphi(2x - n)$. Using the Haar scaling function, we get the Haar wavelet function as

$$\psi(x) = \begin{cases} 1 & 0 \leq x < 0.5 \\ -1 & 0.5 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

3.1 Wavelet series

Any function $f(x) \in L^2(\mathbb{R})$ can be expressed as a series summation of scaling functions and wavelet functions as

$$f(x) = \sum_s a_{r_0,s} \varphi_{r_0,s}(x) + \sum_{r=r_0}^{\infty} \sum_s b_{r,s} \psi_{r,s}(x)$$

The first term of the expansion involving the scaling functions provide approximations to $f(x)$ at scale r_0 and the second term of expansion involving the wavelet functions add details to the approximation at r_0 and its higher scales. If the expansion functions form an orthonormal basis, which is often the case, the coefficients can be calculated as

$$a_{r_0,s} = \int f(x) \varphi_{r_0,s}(x) dx \quad b_{r_0,s} = \int f(x) \psi_{r_0,s}(x) dx$$

4 Subband Coding

4.1 Two-band Analysis

Consider a band-limited 1D signal of cutoff frequency $\omega_c = \pi$, and sampled at exactly twice that frequency, $\omega_s = 2\pi$. Pass this through an LPF with upper cutoff frequency $\omega_0 = \pi/2$ and impulse response $h_0(n)$, and also an HPF with lower cutoff frequency $\omega_1 = \pi/2$ and impulse response $h_1(n)$.

If we perform sub-band analysis on N samples of an input sequence for $n = 0, \dots, N-1$, both the low-frequency and high-frequency analysis filter outputs will have N samples. However, it should be noted that since the bandwidth of the signal at each of the analysis filter outputs is only one-half of the original signal, the analysis filter outputs can be sampled at half the original Nyquist rate. In other words, half of the samples at the analysis filter outputs are redundant and hence every one out of two consecutive samples at the filter output can be dropped. Therefore, the analysis filter outputs are downsampled by a factor of two.

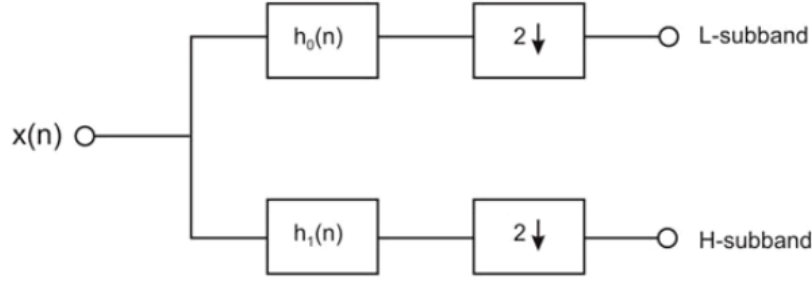


Figure 1: Two-band analysis filters

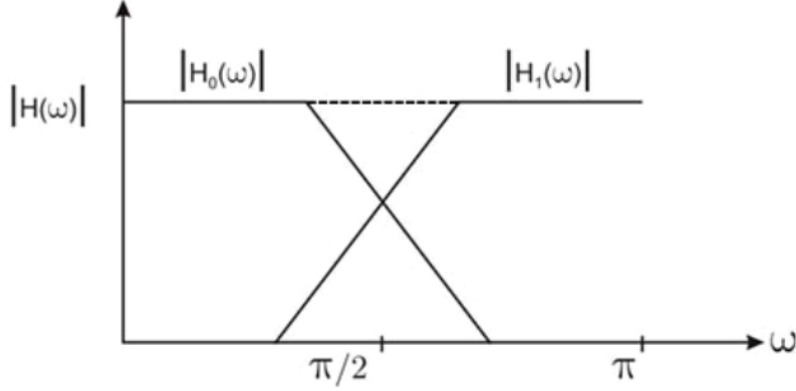


Figure 2: Spectral Components of the two-band analysis filters

4.2 Two-Band Synthesis

The low-pass and high-pass filtered sub-bands are first up-sampled by a factor of two, so that each of the sub-band filters, having impulse responses $g_0(n)$ and $g_1(n)$ respectively for low and high sub-bands, generate N samples for an input sequence of length N samples. The outputs of these two filters are added to generate the reconstructed sequence $\hat{x}(n)$ for $n = 0, \dots, N - 1$.

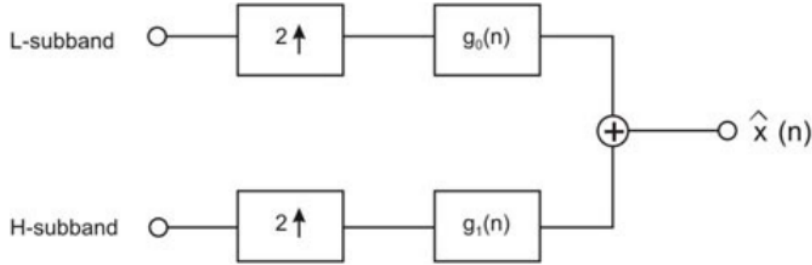


Figure 3: Two-Band Synthesis Filters

4.3 Perfect Reconstruction

For a perfect reconstruction we want $x(n) = \hat{x}(n)$ for $n = 0, \dots, N - 1$. Since $\hat{x}(n)$ depends on $g_i(n)$ and $h_i(n)$, these must satisfy some constraint.

z-transform of down-sampled sequence. The z-transform of the original sequence $x(n)$ is given by

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

Down-sampling $x(n)$ by a factor of two preserves every alternate samples of the original sequence, i.e. $x(2n)$ for $n = 0, 1, \dots$. Its z-transform is therefore given by

$$\begin{aligned}
X_{down}(z) &= \sum_{n=-\infty}^{\infty} x(2n)z^{-n} \\
&= x(0)z^{-0} + x(2)z^{-1} + \dots \\
&= \frac{1}{2}[x(0)z^{-0} + x(1)z^{-1/2} + x(2)z^{-1} + \dots] + \frac{1}{2}[x(0)z^{-0} - x(1)z^{-1/2} + x(2)z^{-1} - \dots] \\
&= \frac{1}{2}[X(z^{1/2}) + X(-z^{1/2})]
\end{aligned} \tag{1}$$

z-transform of up-sampled sequence. In the process of up-sampling, zeros are added for every odd sample and thus, up-sampling a given sequence $x(n)$ for $n = 0, 1, \dots$ can be expressed in terms of the samples of given sequence as

$$x_{up}(n) = \begin{cases} x(n/2) & n = 0, 2, 4, \dots \\ 0 & \text{otherwise} \end{cases}$$

The z-transform of the up-sampled sequence may be expressed as

$$\begin{aligned}
X_{up}(z) &= x(0)z^{-0} + x(1)z^{-2} + x(2)z^{-4} + \dots \\
&= X(z^2)
\end{aligned} \tag{2}$$

Without using any analysis or synthesis filter, if we simply cascade a down-sampler and an up-sampler, then the z-transform of the reconstructed sequence is $\hat{X}(z) = \frac{1}{2}[X(z) + X(-z)]$. With this background, and a bit of rearrangement, we conclude that the reconstructed signal has the z-transform of the form

$$\hat{X}(z) = \frac{1}{2}[G_0(z)H_0(z) + G_1(z)H_1(z)]X(z) + \frac{1}{2}[G_0(z)H_0(-z) + G_1(z)H_1(-z)]X(-z)$$

Thus, for perfect reconstruction and therefore to have $X(z) = \hat{X}(z)$, we want

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2$$

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0$$

The $X(-z)$ part represents the aliased part of the signal. Therefore the first requirement ensures no amplitude distortion and the second ensures no aliasing. Then,

$$\begin{bmatrix} G_0(z) & G_1(z) \end{bmatrix} \begin{bmatrix} H_0(z) & H_0(-z) \\ H_1(z) & H_1(-z) \end{bmatrix} = \begin{bmatrix} 2 & 0 \end{bmatrix}$$

If we let $\mathbf{H}_m(z) = \begin{bmatrix} H_0(z) & H_0(-z) \\ H_1(z) & H_1(-z) \end{bmatrix}$ be the analysis modulation matrix, then we get $\begin{bmatrix} G_0(z) & G_1(z) \end{bmatrix} \mathbf{H}_m(z) = \begin{bmatrix} 2 & 0 \end{bmatrix}$.

Assuming that $\mathbf{H}_m(z)$ is non-singular, we take the transpose of this equation and premultiply it with $(\mathbf{H}_m(z)^T)^{-1}$ to obtain

$$\begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix} = \frac{2}{\det(\mathbf{H}_m(z))} \begin{bmatrix} H_1(-z) \\ -H_0(-z) \end{bmatrix}$$

The above equation reveals that the analysis and the synthesis filters are cross-modulated. For Finite Impulse Response (FIR) filters, $\det(\mathbf{H}_m(z))$ is a pure delay and is equal to $\det(\mathbf{H}_m(z)) = \alpha z^{-(2k+1)}$. Neglecting the delay, letting $\alpha = 2$ and taking inverse transforms, we obtain

$$g_0(n) = (-1)^n h_1(n)$$

$$g_1(n) = (-1)^{n+1} h_0(n)$$

If $\alpha = -2$ the equations are sign-reversed, i.e.

$$g_0(n) = (-1)^{n+1} h_1(n)$$

$$g_1(n) = (-1)^n h_0(n)$$

More Reading: Bi-orthogonality of analysis and synthesis filters.

5 Subband Decomposition of Images

The idea of subband coding can be extended for two-dimensional signals, if the one-dimensional filters used for analysis and synthesis can be used as two-dimensional separable filters. Images having discrete set of samples $s(n_1, n_2)$ are first analyzed into low -frequency and high-frequency subbands through FIR analysis filters along the n_1 -direction, i.e. along the rows (vertically) and then downsampled by factors of 2. Each of the resulting subbands are then analyzed into the low and high-frequency subbands along the n_2 -direction, i.e. along the columns (horizontally) and we thus obtain four subbands, each of which is further sub-sampled by a factor of two.

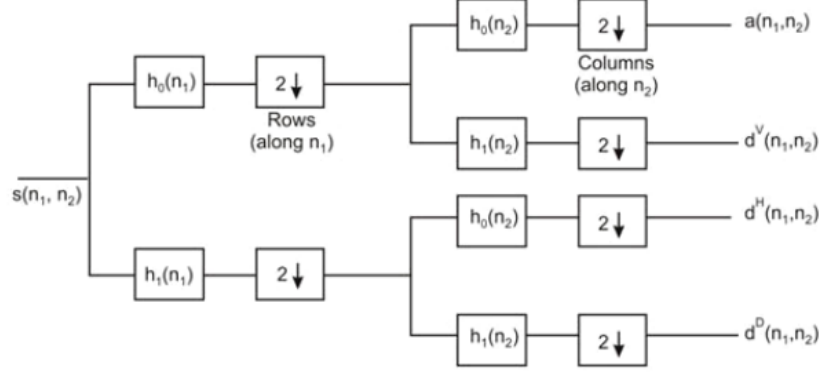


Figure 4: Four-Band analysis of images

The resulting subband outputs $a(n_1, n_2)$, $d^V(n_1, n_2)$, $d^H(n_1, n_2)$ and $d^D(n_1, n_2)$ contain the approximation, the vertical details, the horizontal details and the diagonal details respectively. In terms of resulting images, these subbands are generally expressed as LL, LH, HL and HH, where the first letter indicates the filter applied in the horizontal direction (“L” for low-pass and “H” for high-pass) and the second letter indicates the filter applied in the vertical direction.

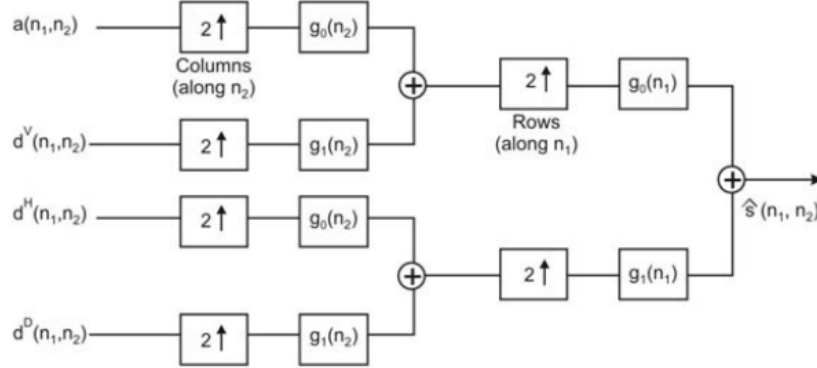


Figure 5: Four-Band Synthesis

6 Discrete Wavelet Transforms

Instead of considering a continuous signal, if we consider a discrete sequence $s(n)$ in equation 3.1, defined for $n = 0, 1, 2, \dots$, the resulting coefficients in the series expansion are called the Discrete Wavelet Transforms (DWT) of $s(n)$. For the discrete case, we can find them as

$$W_\varphi(j_0, k) = \frac{1}{\sqrt{M}} \sum_n s(n) \varphi_{j_0, k}(n)$$

and

$$W_\psi(j, k) = \frac{1}{\sqrt{M}} \sum_n s(n) \psi_{j,k}(n)$$

where $j \geq j_0$ and $s(n), \varphi_{j_0,k}(n)$ and $\psi_{j,k}(n)$ are functions of discrete $n = 0, 1, \dots, M-1$. The corresponding Inverse Discrete Wavelet Transform (IDWT) to express the discrete signal in terms of the wavelet coefficients can be written as

$$s(n) = \frac{1}{\sqrt{M}} \sum_k W_\varphi(j_0, k) \varphi_{j_0,k}(n) + \sum_{j=0}^{\infty} \sum_k W_\varphi(j, k) \psi_{j,k}(n)$$

Normally, we let $j_0 = 0$ and select M to be a power of 2 $M = 2^J$, so that the summations are performed over $j = 0, 1, \dots, J-1$ and $k = 0, 1, 2, \dots, 2^{j-1}$.

6.1 DWT and IDWT through subband analysis and synthesis

We first substitute the definition of the wavelet functions $\psi_{j,k}(n)$ to get:

$$W_\psi(j, k) = \frac{1}{\sqrt{M}} \sum_n s(n) 2^{j/2} \psi(2^j n - k)$$

But we also know that the wavelet function can be expressed as a linear combination of scaling functions: $\psi(n) = \sum_r h_\psi(r) \sqrt{2} \varphi(2n - r)$. If we scale n by 2^j , translate it by k and let $m = 2k + r$, we obtain

$$\begin{aligned} \psi(2^j n - k) &= \sum_r h_\psi(r) \sqrt{2} \varphi(2(2^j n - k) - r) \\ &= \sum_m h_\psi(m - 2k) \sqrt{2} \varphi(2^{j+1} n - m) \end{aligned} \quad (3)$$

Substituting the above expression in our wavelet coefficient equation we get

$$\begin{aligned} W_\psi(j, k) &= \frac{1}{\sqrt{M}} \sum_n s(n) 2^{j/2} \left[\sum_m h_\psi(m - 2k) \sqrt{2} \varphi(2^{j+1} n - m) \right] \\ W_\psi(j, k) &= \frac{1}{\sqrt{M}} \sum_m h_\psi(m - 2k) \left[\sum_n s(n) 2^{(j+1)/2} \varphi(2^{j+1} n - m) \right] \\ W_\psi(j, k) &= \frac{1}{\sqrt{M}} \sum_m h_\psi(m - 2k) \left[\sum_n s(n) \varphi_{j+1,k}(n) \right] = \sum_m h_\psi(m - 2k) W_\varphi(j+1, k) \end{aligned}$$

where W_φ is the DWT for the approximation coefficient. Similarly, if we start from the other equation we can get

$$\begin{aligned} W_\psi(j, k) &= \frac{1}{\sqrt{M}} \sum_m h_\psi(m - 2k) \left[\sum_n s(n) 2^{(j+1)/2} \varphi(2^{j+1} n - m) \right] \\ W_\varphi(j, k) &= \sum_m h_\varphi(m - 2k) W_\varphi(j+1, k) \end{aligned}$$

Thus we have a relationship between the DWT coefficients at adjacent scales: both $W_\psi(j, k)$ and $W_\phi(j, k)$ can be found by convolving the approximation coefficient $W_\varphi(j+1, k)$ with $h_\psi(-n)$ and $h_\phi(-n)$ respectively and then subsampling the convolved output by a factor of 2. This is exactly subband analysis, with $h_0(n) = h_\varphi(-n)$ and $h_1(n) = h_\psi(-n)$!

6.2 2-D DWT

This analysis of 2-D signals requires the use of following two-dimensional filter functions through the multiplication of separable scaling and wavelet functions in n_1 (horizontal) and n_2 (vertical) directions.

$$\varphi(n_1, n_2) = \varphi(n_1) \varphi(n_2)$$

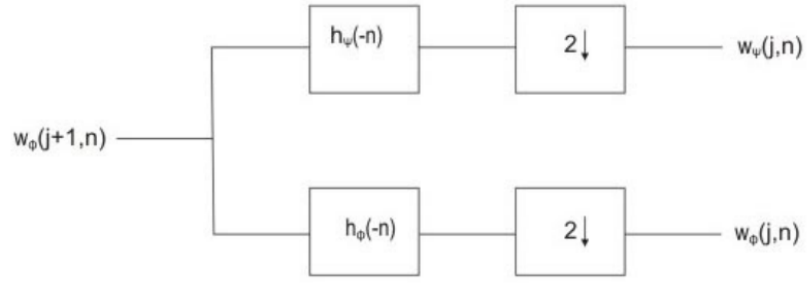


Figure 6: One-stage analysis of DWT coefficients

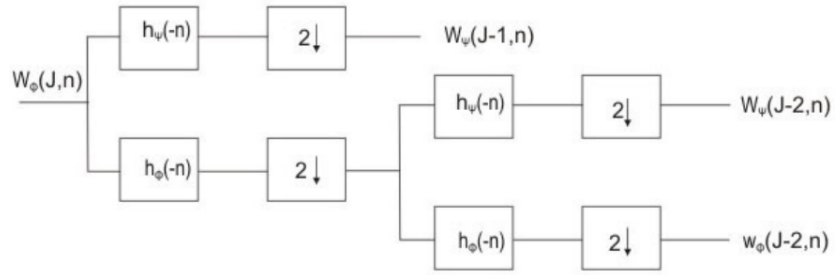


Figure 7: Two-stage analysis of DWT coefficients

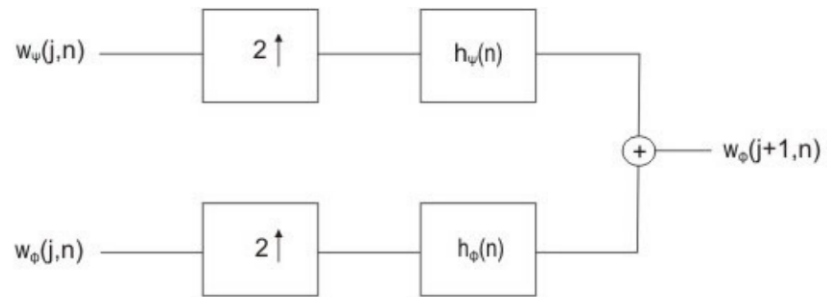


Figure 8: One-stage synthesis of DWT coefficients

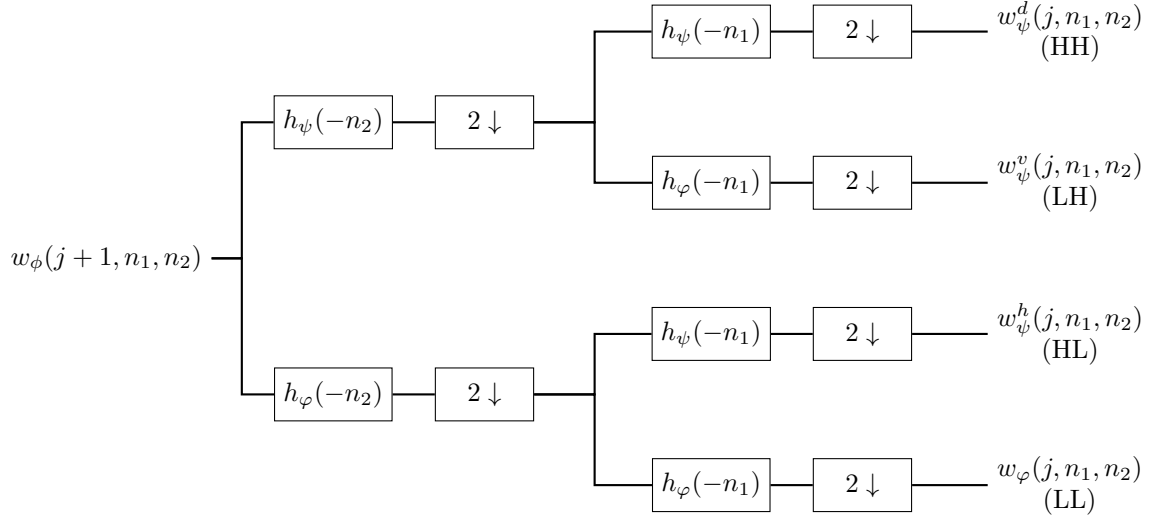


Figure 9: Two-dimensional wavelet decomposition filter bank

$$\begin{aligned}\psi^H(n_1, n_2) &= \psi(n_1)\varphi(n_2) \\ \psi^V(n_1, n_2) &= \varphi(n_1)\psi(n_2) \\ \psi^D(n_1, n_2) &= \psi(n_1)\psi(n_2)\end{aligned}$$

It may be noted that the synthesis filter banks along the rows and columns are associated with an upsampling by a factor of two so that the reconstructed image can be shown at the original resolution. The synthesis filter banks therefore perform the IDWT, which is also lossless, like the DWT.

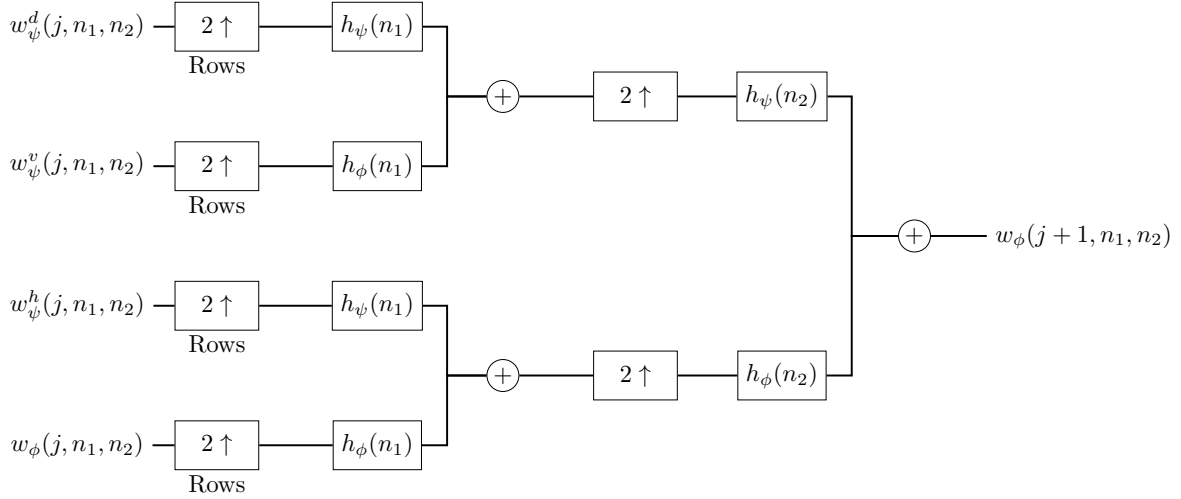


Figure 10: Two-dimensional synthesis filtering for image reconstruction

6.3 Applying 2D DWT to images

The two-dimensional DWT of an image function $s(n_1, n_2)$ of size $N_1 \times N_2$ may be expressed as

$$W_\varphi(j_0, k_1, k_2) = \frac{1}{\sqrt{N_1 N_2}} \sum_{n_1=0}^{N_1-1} \sum_{n_2=0}^{N_2-1} s(n_1, n_2) \psi_{j_0, k_1, k_2}(n_1, n_2) \quad (4)$$

$$W_\varphi^i(j_0, k_1, k_2) = \frac{1}{\sqrt{N_1 N_2}} \sum_{n_1=0}^{N_1-1} \sum_{n_2=0}^{N_2-1} s(n_1, n_2) \psi_{j_0, k_1, k_2}^i(n_1, n_2) \quad (5)$$

where $i \in \{H, V, D\}$ indicate the directional index of the wavelet function. As in one-dimensional case, j_0 represents any starting scale, which may be treated as $j_0 = 0$. Then the image function $s(n_1, n_2)$ may be obtained as the IDWT:

$$s(n_1, n_2) = \frac{1}{\sqrt{N_1 N_2}} \sum_{k_1} \sum_{k_2} W_\phi(j_0, k_1, k_2) \phi_{j_0, k_1, k_2}(n_1, n_2) \quad (6)$$

$$+ \frac{1}{\sqrt{N_1 N_2}} \sum_{i=H, V, D} \sum_{j=0}^J \sum_{k_1} \sum_{k_2} W_\psi^i(j, k_1, k_2) \psi_{j, k_1, k_2}^i(n_1, n_2) \quad (7)$$

6.4 DWT based still image compression system

It may be noted from the results of DWT applied over the images that mostly the LL subband at the highest level of decomposition has significant information content and all other subbands have less significant content. This is expected, since all natural images are rich in low frequency information content, as compared to the high-frequency content and the results demonstrate excellent energy compaction properties of DWT. This property can be very effectively utilized to achieve image compression.

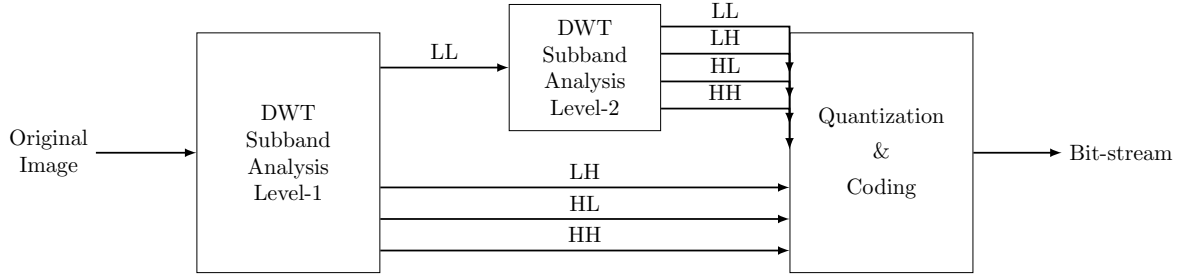


Figure 11: A two-level DWT-based image compression system

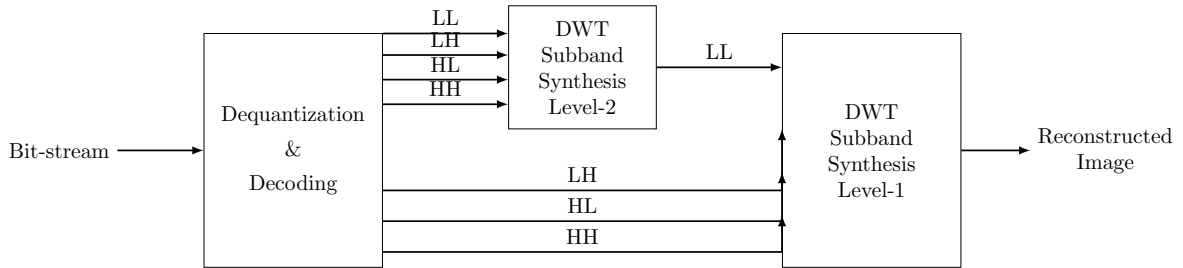


Figure 12: A two-level image decoding system based on DWT